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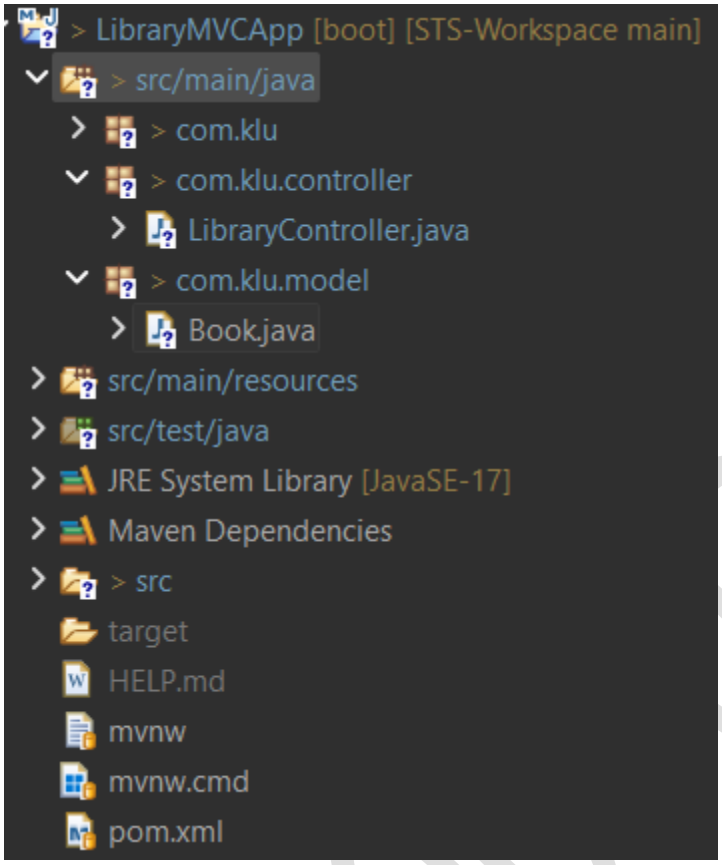
SECTION-59
COURSE: FULL STACK APPLICATION DEVELOPMENT



LibraryMVCAppApplication.java:

```
package com.klu;


import org.springframework.boot.SpringApplication;

import org.springframework.boot.autoconfigure.SpringBootApplication;


@SpringBootApplication
```

public class LibraryMvcAppApplication {

```java
package klu.Spring_Boot_Demo;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class LibraryMVCAppApplication {

    public static void main(String[] args) {
        SpringApplication.run(LibraryMVCAppApplication.class, args);
    }

}
```

```java
        public static void main(String[] args) {
                SpringApplication.run(LibraryMvcAppApplication.class,  args);
        }


}
```

LibraryController.java:

```java
package com.klu.controller;


import com.klu.model.Book;
import org.springframework.web.bind.annotation.*;


import java.util.ArrayList;
import java.util.List;


@RestController
public class LibraryController {


   private List<Book> bookList = new ArrayList<>();
```

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```
@GetMapping("/welcome")
public String welcome() {
    return "Welcome to KLU Online Library System!";
}


@GetMapping("/count")
public int getCount() {
    return 150;
}


@GetMapping("/price")
public double getPrice() {
    return 499.99;
}


@GetMapping("/books")
public List<String> getBooks() {
    List<String> titles = new ArrayList<>();
    titles.add("Java Programming");
    titles.add("Spring Boot Guide");
    titles.add("Data Structures");
    return titles;
}


@GetMapping("/books/{id}")
```

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```
    public String getBookById(@PathVariable int id) {
        return "Details of Book with ID: " + id;
    }


    @GetMapping("/search")
    public String searchBook(@RequestParam String title) {
        return "Searching for book titled: " + title;
    }


    @GetMapping("/author/{name}")
    public String getAuthor(@PathVariable String name) {
        return "Books written by Author: " + name;
    }


    @PostMapping("/addbook")
    public String addBook(@RequestBody Book book) {
        bookList.add(book);
        return "Book added successfully!";
    }


    @GetMapping("/viewbooks")
    public List<Book> viewBooks() {
        return bookList;
    }
}
```

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Book.java:

```
public class Book {

    private int id;
    private String title;
    private String author;

}
```

```
package com.klu.model;


public class Book {


    private int id;

    private String title;

    private String author;

    private double price;


    public Book() {

    }


    public Book(int id, String title, String author, double price) {

        this.id = id;

        this.title = title;

        this.author = author;

        this.price = price;

    }


    public int getId() {

        return id;

    }


    public void setId(int id) {

        this.id = id;
```

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```
    }

    public String getTitle() {
        return title;
    }

    public void setTitle(String title) {
        this.title = title;
    }

    public String getAuthor() {
        return author;
    }

    public void setAuthor(String author) {
        this.author = author;
    }

    public double getPrice() {
        return price;
    }

    public void setPrice(double price) {
        this.price = price;
    }
}
```

OUTPUT:

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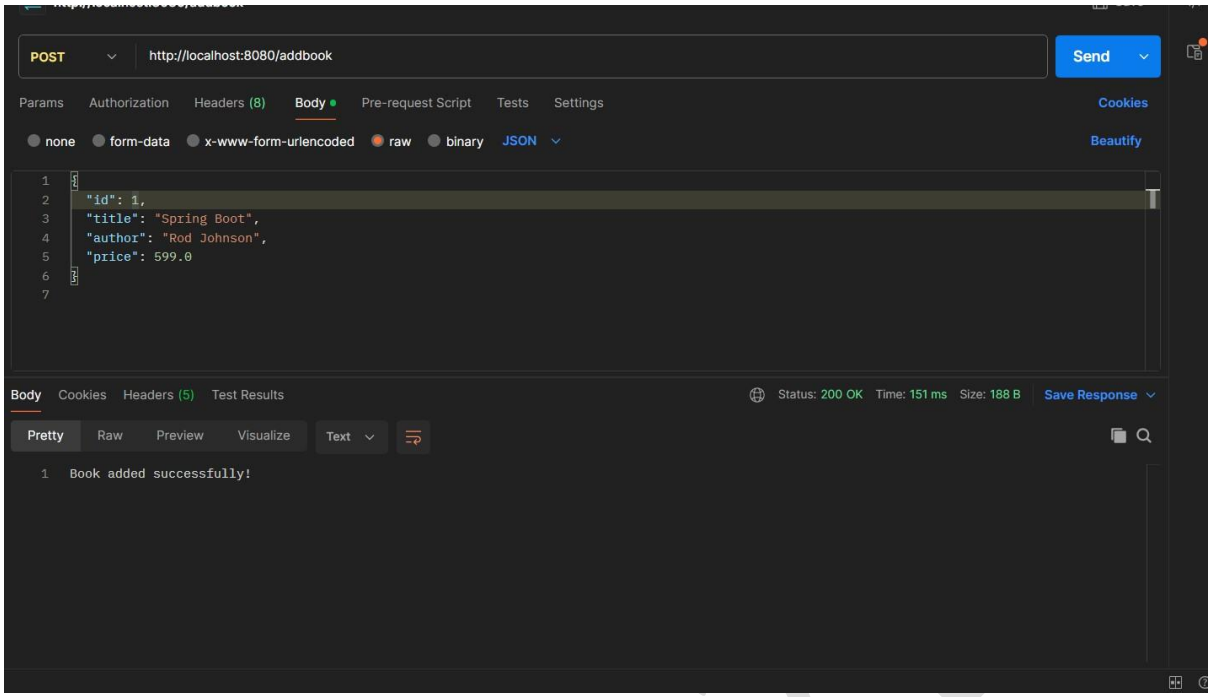

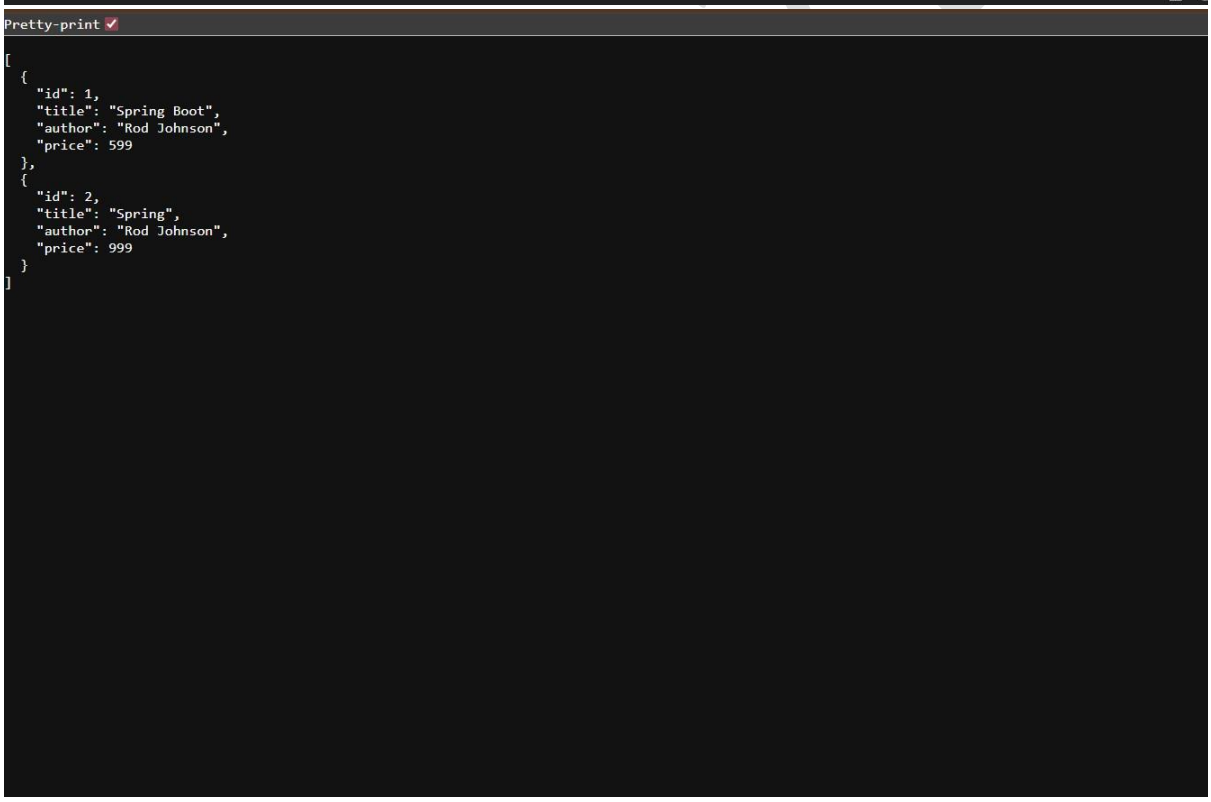

VIVA QUESTIONS:

1. **What is the role of @RestController in Spring MVC?**

@RestController is used to create REST APIs.
It combines @Controller + @ResponseBody and returns data (JSON) directly instead of a view.

2. **What is the difference between @GetMapping and @PostMapping?**

   - `@GetMapping` → Used to **retrieve data** from the server.
   - `@PostMapping` → Used to **send/create data** on the server.

3. **How does @PathVariable help in handling dynamic URLs?**
   @PathVariable is used to get values from the URL.
   Example:
   /courses/1 → 1 is captured using @PathVariable.
4. **When do we use @RequestParam?**
   @RequestParam is used to get values from query parameters.
   Example:
   /search?title=Java
5. **Can one controller contain multiple request-mapped methods?**
   Yes
   One controller can have multiple methods with different mappings like @GetMapping, @PostMapping, etc.

Git Hub link: https://github.com/hemalatharachakonda/2400030211-Skill_experiments.git

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